

2. Literature Review

2.1 Review focus

Review existing approaches in:

1. Residential property management systems
2. Utility billing and shared-cost allocation systems
3. Mobile-first admin systems
4. Offline-capable applications
5. Alert/notification systems in community platforms

2.2 Literature review content :

Include:

Existing building/property management solutions

Common limitations in local contexts like Egypt

Importance of transparent utility distribution

Benefits of mobile apps for resident engagement

Need for offline-first design where connectivity is inconsistent

Role of reporting and alerts in operational efficiency

2.3 Gap identified

Most generic property management systems focus on rent, leasing, and landlord workflows, while your system focuses more specifically on:

shared building utilities

admin-approved resident access

apartment-level distribution logic

localized building operations

monthly resident-facing reports

That becomes your research and design gap.